

Privacy-Utility Tradeoffs in Healthcare AI: A Comparative Analysis of De-Identification Techniques on Machine Learning Model Performance

Tai Chou-Kudu

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Instructor: Professor Nasrin Khansari

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CUNY School of Professional Studies

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Abstract

Healthcare organizations deploying machine learning models face a fundamental tension between patient privacy and data utility. De-identification is the most widely adopted approach to managing this tension, yet little empirical evidence exists on how different de-identification techniques affect model performance across clinical prediction tasks. This study applies k-anonymity and data-level differential privacy to the MIMIC-III critical care database and evaluates their impact on two prediction tasks: 30-day hospital readmission and in-hospital mortality. For each technique, a range of privacy parameter values is tested ($k=5$ through $k=50$ for k-anonymity and $\epsilon=0.1$ through $\epsilon=10$ for differential privacy), and a logistic regression model is trained and evaluated on each resulting dataset. Model performance is measured using AUROC and F1 score; re-identification risk is assessed using a structural uniqueness check (k-anonymity) and a membership inference attack (differential privacy). K-anonymity produced surprisingly minimal performance degradation across all k values, while differential privacy caused substantial accuracy loss at strict privacy settings, with mortality AUROC dropping from 0.691 to 0.443 at $\epsilon=0.1$. Performance degradation was consistent across both prediction tasks, suggesting that de-identification technique matters more than the clinical outcome being predicted for admission-level feature sets. The findings provide practical guidance for data scientists and compliance teams on selecting de-identification approaches that balance patient privacy with clinical utility.

Introduction

Healthcare organizations are under growing pressure to adopt AI-driven clinical tools while simultaneously protecting patient privacy. Machine learning models for predicting outcomes like hospital readmission and in-hospital mortality require access to detailed patient records, including demographics, lab values, diagnoses, and medications. The same richness that makes this data clinically valuable also makes it a target for re-identification attacks. Under HIPAA, covered entities may share de-identified data without the use and disclosure restrictions that apply to protected health information. However, the two recognized methods, Safe Harbor and Expert Determination, were designed for data release rather than the iterative, feature-dependent demands of ML model training.

In practice, both k-anonymity and data-level differential privacy are applied to the dataset after collection and before model training, meaning privacy decisions made upstream directly shape what the model can learn downstream. Applying stronger de-identification tends to degrade model accuracy; applying weaker protections leaves patients exposed to re-identification risk. Existing studies examine k-anonymity and differential privacy largely in isolation, and comparisons across clinically distinct prediction tasks on the same dataset remain limited.

This study addresses that gap. Specifically, this study makes three contributions:

- An empirical, head-to-head comparison of k-anonymity and differential privacy on the same clinical dataset (MIMIC-III), rather than evaluating each technique in isolation.
- A demonstration that de-identification technique matters more than the clinical outcome being predicted: readmission prediction (6.25% positive rate, discharge outcome) and in-hospital mortality (10.86% positive rate, in-hospital outcome) responded nearly identically to both techniques when models were built on admission-level features.

- A practical decision framework for compliance teams and data scientists to select a de-identification approach based on their prediction task and acceptable accuracy loss.

Literature Review

AI adoption in healthcare creates a direct tension between clinical utility and patient privacy.

Chustecki (2024) found in a synthesis of 44 peer-reviewed studies that AI systems show measurable advantages in disease prediction and resource allocation while simultaneously introducing risks around data opacity and privacy exposure. Chang et al. (2025) found that only 31% of hospitals monitor AI for privacy compliance and nearly half lack formal approval processes, highlighting the need for empirical guidance.

The HIPAA Privacy Rule provides two legal pathways for de-identifying protected health information: the Safe Harbor method, which requires removal of 18 specific identifier categories, and the Expert Determination method, which requires a qualified expert to certify that re-identification risk is very small (HHS, 2012). While Safe Harbor is widely used for its operational simplicity, requiring the removal of specific date elements beyond the year makes it difficult to compute time-sensitive outcomes such as 30-day readmission windows. AI systems further complicate both methods by enabling re-identification through the Mosaic Effect.

Research has shown that a combination of date of birth, sex, and five-digit ZIP code can uniquely identify over 50% of U.S. residents, whereas using year of birth, sex, and three-digit ZIP code reduces this to under 0.1%, illustrating why quasi-identifier selection directly affects the privacy-utility balance (Sweeney, 2000).

Prior empirical work has largely examined de-identification techniques in isolation. K-anonymity, formalized by Sweeney (2002), requires each record to be indistinguishable from at

least $k-1$ others through generalization and suppression. Slijepčević et al. (2021) found that classification performance degrades as k strengthens, with the degree of loss depending on the correlation between generalized features and the target variable. Im et al. (2024) confirmed measurable accuracy loss in clinical outcome prediction, with magnitude varying by task and quasi-identifier configuration. Differential privacy, formalized by Dwork and Roth (2014), adds calibrated noise with a formal mathematical guarantee parameterized by epsilon. A scoping review by Tayebi Arasteh et al. (2026) in *npj Digital Medicine* synthesizing 74 studies found that moderate privacy budgets (epsilon ≈ 10) maintain acceptable performance in medical applications, while strict settings (epsilon ≈ 1) often cause substantial accuracy loss and can widen performance gaps across demographic subgroups.

Thirty-day readmission and in-hospital mortality have both been studied extensively as benchmark prediction tasks on MIMIC-III. Prior work on readmission has used increasingly rich feature sets: Assaf and Jayousi (2023) showed that adding text embeddings from discharge notes to structured features improved performance, and LSTM-based approaches on Medicare patients have reported AUROCs in the 0.70–0.78 range using sequential visit data (Pishgar et al., 2024). Mortality prediction tends to perform better; a 2025 meta-analysis across 40 ICU studies found a pooled AUROC of 0.83 across logistic regression, random forest, and XGBoost models (Sun et al., 2025). Direct comparisons of de-identification techniques across both tasks on the same dataset remain limited, which this study addresses.

Data and Methods

This study uses the Medical Information Mart for Intensive Care III (MIMIC-III) database, a critical care database developed by Johnson et al. (2016) at MIT and Beth Israel Deaconess

Medical Center. MIMIC-III contains de-identified clinical records for patients admitted to the ICU at Beth Israel Deaconess Medical Center between 2001 and 2012. Access requires completion of human subjects research training through the CITI program and execution of a Data Use Agreement with PhysioNet, both of which have been completed prior to this analysis. The analysis dataset is constructed by joining three MIMIC-III tables: ADMISSIONS, PATIENTS, and LABEVENTS. The study population is restricted to adult patients between 18 and 89 years of age at the time of admission, excluding patients whose ages could not be reliably computed due to MIMIC-III's date-shifting protocol for patients over 89, which produces implausibly large age values. After applying this filter, the working dataset contains 48,180 admissions.

Nine features are used as model predictors: age at admission, gender, ethnicity, admission type, creatinine, blood urea nitrogen (BUN), bicarbonate, white blood cell count (WBC), and sodium. Lactate was excluded due to 32.2% missingness; without clinical domain expertise to evaluate whether imputation would be appropriate, exclusion was the more defensible choice. The remaining lab features had low missingness rates (all below 2%), and the analysis applies median imputation; a threshold of approximately 5% is generally considered to have negligible impact on model estimates (Jakobsen et al., 2017). Label encoding converts categorical variables to numeric indicators.

Two clinical prediction tasks are evaluated. In-hospital mortality is encoded in MIMIC-III as HOSPITAL_EXPIRE_FLAG, where a value of 1 indicates the patient died before discharge. Thirty-day readmission is defined as any new admission within 30 days of discharge, excluding patients who died during the index admission. In the working dataset of 48,180 admissions, the mortality rate is 10.86% and the 30-day readmission rate among surviving patients is 6.25%.

Both are binary classification problems with class imbalance. This study applies class weighting using scikit-learn's balanced setting, which increases the penalty for misclassifying the minority class proportionally to how underrepresented it is. Oversampling was not used because generating synthetic records prior to de-identification would obscure whether performance changes were attributable to the privacy transformation or to the artificial inflation of the minority class. Because class weighting shifts the model's decision boundary toward the minority class, predicted probabilities no longer reflect the true class distribution in the data. This study therefore uses AUROC as the primary evaluation metric for its threshold-independence; F1 is reported for comparison across privacy conditions only.

The modeling pipeline proceeds in five stages for each de-identification configuration:

- Raw dataset extracted from BigQuery and preprocessed (imputation, encoding).
- De-identification transformation applied (k-anonymity or differential privacy at the specified parameter value).
- Logistic regression model trained using scikit-learn with `class_weight='balanced'`.
- Both AUROC and F1 are evaluated using 5-fold stratified cross-validation on the full dataset, with class balance preserved across folds.
- AUROC and F1 reported as means across the five held-out folds.

A baseline model trained on the raw, unmodified dataset is included as the performance ceiling against which all de-identified configurations are compared. This pipeline is executed for each of the 16 de-identification configurations (four k values and four epsilon values, each applied to two prediction tasks), plus two baseline evaluations on the raw dataset.

This study uses logistic regression deliberately because isolating the effect of de-identification on model performance requires a simple, interpretable model. If the model were more complex, it

would be harder to tell whether performance changes came from the privacy transformation or from the model itself.

Two de-identification techniques are applied. This study implements k-anonymity via manual generalization: age is binned into decade ranges, and ethnicity is grouped from 41 distinct values into six broader categories (White, Black/African American, Hispanic/Latino, Asian, Other, and Unknown). Admission type is left as-is given its limited number of values. This implementation intentionally scopes k-anonymity to demographic quasi-identifiers (age and ethnicity) while leaving continuous clinical laboratory features unaltered. In a practical deployment, highly specific clinical profiles across multiple continuous lab values could potentially serve as quasi-identifiers for re-identification attacks. However, limiting the transformation to demographic features was a deliberate design choice to cleanly isolate how categorical generalization affects downstream model utility relative to global, data-level noise injection on continuous features.

The pycanon library measures the proportion of records satisfying the k-anonymity threshold for each value of k tested, reported as the compliance rate in the results. It is worth noting that while formal k-anonymity requires 100% of records to reside within an equivalence class of size $\geq k$, this study treats the metric evaluated by pycanon as an assessment of data-level compliance rather than a strict enforcement filter. Records that failed to satisfy the designated k-threshold were intentionally retained in the modeling pipeline to preserve a consistent sample size ($n = 48,180$) across all experimental configurations. Consequently, the compliance metric reported in the results reflects the percentage of the dataset that naturally satisfies the privacy threshold under the applied generalization rules, serving as an indicator of structural vulnerability rather than absolute compliance execution.

The same grouped ethnicity encoding is applied to the baseline model and all experimental configurations to ensure a consistent feature representation across all comparisons, isolating the effect of the privacy transformation itself. Four values of k are tested: $k=5$, $k=10$, $k=25$, and $k=50$. This study implements data-level differential privacy using IBM's `diffprivlib` library, which injects calibrated Laplace noise into continuous features (age, creatinine, BUN, bicarbonate, WBC, sodium). `diffprivlib` adds noise only to continuous features, as the Laplace mechanism is designed for numeric data. Four epsilon values are tested: $\epsilon=0.1$, $\epsilon=1$, $\epsilon=5$, and $\epsilon=10$.

The study assesses re-identification risk for each configuration using two measures. For k -anonymity, a structural uniqueness check calculates the proportion of records remaining uniquely identifiable after de-identification. For differential privacy, a membership inference attack simulates an adversary attempting to determine if a specific record was used in training. The attack uses a 70/30 stratified train/test split: the model is trained on 70% of the data, and attack accuracy is measured by how well the model's confidence scores distinguish training records from held-out records. Together, these provide both a data-level and model-level view of residual privacy risk. All code is publicly available at <https://github.com/tai-chou-kudu/healthcare-privacy-ml>.

Results

The results section reports findings for each de-identification technique across both prediction tasks. The dataset with grouped ethnicity encoding and no privacy transformation applied serves as the baseline, and all de-identified configurations are measured against it. Table 1 covers k -

anonymity across four values of k ; Table 2 covers differential privacy across four values of ϵ .

The baseline logistic regression model ($n=48,180$) achieved an AUROC of 0.691 and F1 of 0.277 for in-hospital mortality, and a cross-validated AUROC of 0.592 and F1 of 0.144 for 30-day readmission. The readmission AUROC of 0.592 is on the lower end of what the literature reports for similar structured-feature models, which is expected given the feature set here excludes comorbidity codes, discharge notes, and vitals. The goal was not to optimize readmission prediction but to establish a consistent baseline comparable across privacy conditions. Lower F1 scores relative to AUROC are expected under class imbalance, as F1 is sensitive to the minority class size while AUROC measures ranking performance across all thresholds.

Table 1. K-anonymity results: model performance and re-identification risk by k value.

k	k-Anonymity Compliance Rate (%)	Mortality AUROC	Mortality F1	Readmission AUROC	Readmission F1
Baseline	100%	0.691	0.277	0.592	0.144
k=5	99.7%	0.691	0.275	0.592	0.145
k=10	99.3%	0.691	0.275	0.592	0.145
k=25	98.3%	0.691	0.275	0.592	0.145
k=50	95.7%	0.691	0.275	0.592	0.145

Table 2. Differential privacy results: model performance and membership inference attack accuracy by epsilon value.

Epsilon	MI Attack Acc. Mortality	MI Attack Acc. Readmission	Mortality AUROC	Mortality F1	Readmission AUROC	Readmission F1
Baseline	0.464	0.444	0.691	0.277	0.592	0.144
0.1	35.4%	33.2%	0.443	0.096	0.515	0.084
1	34.4%	32.9%	0.568	0.133	0.500	0.085
5	32.6%	32.0%	0.614	0.099	0.512	0.052
10	31.3%	31.1%	0.662	0.057	0.543	0.035

K-anonymity produced surprisingly minimal performance degradation across all k values. Mortality AUROC held at 0.691 and F1 at 0.275 across k=5 through k=50, compared to a baseline of 0.691 and 0.277 respectively. Readmission AUROC and F1 were similarly stable at 0.592 and 0.145 across all k values. The structural uniqueness check showed that compliance remained high even at stricter k thresholds, dropping from 99.7% at k=5 to 95.7% at k=50. The result suggests that the k-anonymity transformations applied, age generalization and ethnicity grouping, had minimal impact on model utility, with AUROC remaining within 0.001 of baseline across all k values. Continuous lab values such as creatinine, BUN, and bicarbonate were left unmodified by k-anonymity, and these likely carried the majority of predictive signal for both tasks. Differential privacy showed a markedly different pattern. At strict privacy settings (epsilon=0.1), mortality AUROC dropped from 0.691 to 0.443, near or below chance level, and readmission AUROC fell from 0.592 to 0.515. At more permissive settings (epsilon=10), mortality AUROC partially recovered to 0.662 and readmission to 0.543, but neither task approached baseline performance. F1 scores did not change consistently with epsilon and varied irregularly across configurations.

The membership inference attack showed that the baseline model achieved attack accuracies of 0.464 for mortality and 0.444 for readmission, both already below chance level, suggesting the model does not memorize training records under standard conditions. Under differential privacy, lower epsilon should make the model harder to attack. All DP configurations reduced attack accuracy relative to baseline, consistent with this expectation. The ordering within DP configurations was less clear, with epsilon=10 achieving the lowest attack accuracy rather than epsilon=0.1, likely reflecting noise variability in the simplified attack implementation. This further decline into the 31% to 35% range presents a mathematical artifact of the evaluation framework. Because the simplified attack relies heavily on prediction confidence scores, the combination of heavy continuous noise injection and uncalibrated probability thresholds introduced by class weighting systematically shifts the model's prediction signatures. This causes the adversary to consistently under-predict membership on the training set, resulting in an artificial drop significantly below the 0.50 random-chance threshold.

Discussion

The results illuminate the practical tradeoffs healthcare organizations face when choosing between k-anonymity and differential privacy for clinical ML applications. The two techniques differed sharply in how they affected model performance.

K-anonymity had minimal impact on model performance. The transformations applied, age banding and ethnicity grouping, did not affect the continuous lab features that drove prediction for both tasks. Both 30-day readmission and in-hospital mortality showed nearly identical AUROC degradation across all k values, which is expected given both used the same admission-level feature set. The compliance drop from 99.7% at k=5 to 95.7% at k=50 is worth noting:

stronger k requirements left more records in non-compliant groups without improving model utility in any measurable way. Incorporating longitudinal features such as prior admission frequency or lab value trends could potentially have produced a starker difference between tasks, but testing that requires a different study design. Differential privacy worked differently, injecting noise directly into the continuous lab features that k -anonymity left untouched. That is why it caused far more damage to model performance, particularly at low epsilon values where the noise overwhelmed the signal. The membership inference attack confirmed that differential privacy reduces model-level privacy risk, though the baseline model was already resistant to membership inference, suggesting the additional protection comes at a steep utility cost for this dataset.

Choosing between the two techniques comes down to organizational risk priorities. K -anonymity produces a clear, auditable output where every record looks like at least $k-1$ others, making it well suited for regulatory compliance documentation. Differential privacy is more appropriate when the concern is that the trained model itself could be exploited to reconstruct patient information. For readmission prediction, where clinical accuracy matters most, both techniques impose a real cost, and organizations may need to weigh whether weaker privacy settings or alternative approaches such as federated learning or synthetic data generation are worth considering. Federated learning in particular avoids centralizing patient data by training models locally across institutions, though it introduces its own costs including communication overhead, cloud infrastructure for model aggregation, statistical heterogeneity across sites, and governance complexity.

Table 3, below, summarizes a practical decision framework for selecting a de-identification technique based on organizational priorities and acceptable utility loss, grounded in the empirical results of this study.

Organizational Priority	Recommended Technique	Expected AUROC Impact	Key Tradeoff
Regulatory compliance, auditability	K-anonymity ($k=5-10$)	Minimal (<0.001 drop)	Record compliance drops from 99.7% to 95.7% as k increases
Model-level privacy, adversarial protection	Differential privacy ($\epsilon=10$)	Moderate ($\sim 0.03-0.05$ drop)	Membership inference attack accuracy drops to 0.313
Maximum privacy, utility secondary	Differential privacy ($\epsilon=0.1$)	Severe (~ 0.25 drop, near chance)	Model likely not suitable for clinical use at this setting

Table 3. Decision framework for selecting a de-identification technique based on organizational priorities. AUROC impact values are approximate and based on mortality prediction results; readmission results followed the same pattern.

This study has several limitations. First, only logistic regression is evaluated; results may not generalize to more complex model architectures such as random forest, gradient boosting, or neural networks. Second, the privacy metrics differ between techniques (structural uniqueness for k -anonymity and membership inference for differential privacy), which makes direct cross-technique comparison harder to interpret. Third, the single-institution dataset from MIMIC-III limits generalizability to other clinical settings. Fourth, the computational cost of repeated model training across multiple de-identification configurations, while modest here, is worth acknowledging as healthcare AI systems scale. Fifth, the two techniques are not fully symmetric in their application: k -anonymity generalizes both continuous and categorical features, while differential privacy adds noise only to continuous features, leaving categorical quasi-identifiers unchanged. This difference in scope should be considered when interpreting the cross-technique comparison. Sixth, differential privacy results are subject to run-to-run variability due to the

stochastic nature of noise injection; the reported values represent a single experimental run and directional findings are consistent across runs, but exact numbers may differ slightly upon re-execution.

Conclusion

This study set out to answer a practical question: when applying de-identification to clinical data for machine learning, does the choice of technique matter, and does it matter differently depending on the prediction task? The answer is that de-identification technique matters greatly, while the clinical outcome being predicted, at least for admission-level feature sets, may not. K-anonymity preserved model performance almost entirely across all k values, while differential privacy caused severe degradation at strict privacy settings. Both prediction tasks, 30-day readmission and in-hospital mortality, responded nearly identically to both de-identification techniques, suggesting that when models are built on single-admission features, the mechanism of de-identification dominates the clinical nature of the outcome being predicted.

This study's empirical benchmarks give organizations a concrete reference point for evaluating de-identification choices, rather than applying default settings without justification. For organizations prioritizing regulatory compliance and auditability, k -anonymity at $k=5$ preserved nearly all model utility while achieving 99.7% record compliance. For organizations concerned about attempts to exploit a deployed model to extract patient information, differential privacy at $\epsilon=10$ is the better choice. It reduced membership inference attack accuracy meaningfully, though at the cost of 4 to 5 AUROC points relative to baseline. At stricter settings, the utility cost of differential privacy became prohibitive for clinical use.

Future work should extend this comparison to additional model architectures (random forest, gradient boosting, XGBoost, and neural networks) to assess whether tradeoff patterns hold beyond logistic regression. Additional clinical prediction tasks such as length of stay and diagnosis classification would further test generalizability. Future work should also incorporate cross-visit temporal features for readmission prediction, such as prior admission frequency, days since last discharge, and lab value trends across visits, to properly test whether longitudinal feature sets are more sensitive to de-identification than admission-level ones. A more rigorous membership inference evaluation using shadow model approaches, such as that proposed by Shokri et al. (2017), would strengthen the privacy assessment beyond the simplified attack used here. Additionally, replacing uniform age bands with clinically meaningful thresholds, such as grouping by surgical candidacy or mortality risk categories, would test whether smarter generalization changes the privacy-utility tradeoff. Comparisons with federated learning and synthetic data generation would situate de-identification within the broader landscape of privacy-preserving ML.

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